ADITYA GIRISH

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EDUCATION

Bachelor of Technology (B.Tech.), Computer Science, 2022-2026 — **PES University, Bengaluru**

PUBLICATIONS

When Does Structure Help? Graph vs. Tabular Models for Agent-Workflow Cost Prediction
AAAI Conference on Artificial Intelligence (AAAI 2027) — **Under Review — Main Technical Track**
AAAI Press Proceedings · CORE A∗ · Double-blind peer review — *Lead Author*
Investigates when graph neural networks outperform tabular models for agent-workflow cost prediction, demonstrating across 357K real coding-agent rounds that graph models lose when execution costs do not propagate through workflow structure.

MSTRAE: Multi-Scale Temporal Recurrent Autoencoder for Flight Manoeuvre Recognition
Future Technologies Conference (FTC) 2026 — **Accepted — Oct 2026, Berlin**
Springer Lecture Notes in Networks and Systems · Indexed: Web of Science, SCOPUS, INSPEC — *Lead Author*
Trained a multi-scale LSTM autoencoder in PyTorch with HMM-based unsupervised clustering and an LLM-based evaluation framework for zero-shot flight manoeuvre recognition on Cessna 172 telemetry data.

EXPERIENCE

TruEstate — *Bengaluru, India*
AI/ML Engineer — **Jul 2026 - Present**

- Consulting on **voice-AI infrastructure and orchestration** for an outbound cold-calling pipeline — stack selection across telephony, STT, LLM, and TTS, latency budgeting, and turn-taking/barge-in handling — and owning **context curation** that grounds each call in CRM and listing data, so agents open with lead-specific talking points instead of a static script.
- Built a **Meta ad creative pipeline** generating image and video ad variants directly from listing data, replacing manual design turnaround with automated creative production.

Software Engineering Intern - AI — **Jan 2026 - Jul 2026**

- Architected a **customer-facing AI agent** with **tool calling** for user data retrieval and hierarchical vector search for selective context; a single **classification-routed instance** handles sales, service-delivery, and post-sales queries, replacing a three-agent design and cutting deployment overhead. Prototype slated for production release.
- Designed and deployed an **agent-based CMS** on GCP for one-click microsite generation, with embedded Cloudflare deployment, OTP verification, and **CRM endpoints routing every lead from generated sites into the CRM**; cut broker site-build time from **2 hours to 5 minutes** and improved lead quality by **60%**.
- **Presented technical solutions to 500+ brokers across two industry events**, translating complex AI capabilities into clear business value and gathering customer requirements that shaped subsequent product iterations; built the underlying image-generation pipeline (Gemini 2.5 Flash Image) staging interiors across 200+ property listings.
- Discovered, demonstrated, and designed a **2D-to-3D reconstruction pipeline** converting architectural floor plans into interactive 3D models, using a **VLM (Gemini)** for layout element detection and spatial parsing, with **Three.js** rendering.

AI Backend Engineer Intern — **Apr 2025 – May 2025**
Viable Ideas Private Limited — *Bengaluru, India*

- Trained an **LLM-based** inventory prediction model on transaction data from 4 bank sources, reaching **85% classification accuracy in production** and automated data pipelines for multi-source financial data, cutting manual workflows by **70%** and operational costs by 40%.

PROJECTS

QRAG : Quantized Bi-Level Retrieval Augmented Generation

- Developed a memory-efficient RAG algorithm using dual-level quantization, achieving **3-4x memory reduction** and **95.1% recall@10** on the FIQA dataset.
- Benchmarked against FAISS baselines across 57,000 documents; outperformed IVF-PQ by **21.5pp in recall** at comparable memory budgets.

OriginScale : Low-Latency Deterministic Clustering Initialization

- Developed an efficient alternative to traditional K-Means clustering, achieving constant initialization times of just 26ms against the industry standard's average of approximately 3.5s during trials with large datasets up to 200K entries.
- Applied it to a real-time food-delivery dispatch system, achieving **1.68× lower latency** with no loss in assignment quality, prioritizing speed and consistency where response time matters.